Exponential Models

Name _____

Algebra 2 Lesson 6-2 (fast track) Date _____ Period _____

Learning goal: I can rewrite an exponential model to find its growth rate for a different time period, and write an exponential model from two points.

Language goal: I can explain how I found a rate or a model, using the words *growth factor*, *growth rate*, *initial value*, and *Power of a Power rule*.

On the quiz: (Savvas numbers) You will (1) write the model for a town that shrinks 2.4% per year and find its rate per quarter (Example 1); (2) find another point on $y = a \cdot b^x$ from two points (Example 5); (3) bonus: the model for an account compounded monthly (IXL YJW). (4) and (5), interest: full Savvas quiz only.

On the Topic 6 test: Question 2 asks which model shows butterflies growing from 2000 to 2800 in 7 years (Example 1). Question 5 asks for another point on $y = a \cdot b^x$ from the views after 1 and 2 days (Example 5). Question 6 asks which model of a 20% yearly growth shows the monthly rate (Example 1).

Example 1: Rewrite an Exponential Function to Identify a Rate

In 2021, the population of a small town was 8,000. The population is increasing at a rate of 2.5% per year. Rewrite an exponential growth function to find the monthly growth rate.

1. Write a model with the annual rate for the population y, t years after 2021. The **initial value** is 8,000; the **growth rate** is 0.025; the **growth factor** is $1 + 0.025$.

$$y = 8{,}000(1 + 0.025)^t = 8{,}000(1.025)^t$$

2. For a monthly rate, the exponent must be the number of months in t years, $12t$. Multiply the exponent by $\frac{12}{12}$, then use the **Power of a Power rule**, $(b^m)^n = b^{mn}$:

$$\begin{aligned} y &= 8{,}000(1.025)^{\frac{12t}{12}} \\ &= 8{,}000\left(1.025^{\frac{1}{12}}\right)^{12t} \\ &\approx 8{,}000(1.00206)^{12t} \end{aligned}$$

3. The monthly growth factor is about 1.00206, so the monthly growth rate is about $1.00206 - 1 = 0.00206$.

The population is increasing about 0.206% per month.

Common error. Dividing the annual growth rate by 12 ($0.025 \div 12 \approx 0.00208$) does not give the exact monthly growth rate. You must use the Power of a Power rule to find the exact monthly growth rate.

Other periods. For a rate per quarter, multiply the exponent by $\frac{4}{4}$ instead: the quarterly growth factor is $(1 + r)^{\frac{1}{4}}$, not $1 + \frac{r}{4}$.

Say it: The yearly *growth factor* is _____. I used the *Power of a Power rule* to make the exponent _____, so the monthly *growth factor* is about _____ and the monthly *growth rate* is about _____%, because ...

Try It 1: Quarterly rate

The population in a small town is increasing annually by 1.8%. What is the quarterly rate of population increase?

Hint: multiply the exponent by $\frac{4}{4}$, as Example 1 multiplied it by $\frac{12}{12}$.

$y = a\,(_____)^t = a\left(_____^{1/4}\right)^{4t} \approx a\,(_____)^{4t}$

Quarterly growth rate $\approx$ _____ $- 1 =$ _____, or about _____% per quarter.

Example 5: Use Two Points to Find an Exponential Model

Tia knew that the number of instant messages (IM) she sent was growing exponentially. She generated a record of the number of IM she sent each year since 2018. What is an exponential model that describes the data?

Year	IM sent
2022	6,000
2023	6,600

Write an exponential model in the form $y = a \cdot b^x$, with y equal to the number of IM in hundreds and x equal to the number of years since 2018. Use the data to find the values of the constants a and b.

1. 2022 is $x = 4$ with $y = 60$ hundred; 2023 is $x = 5$ with $y = 66$ hundred.
2. The x-values are consecutive integers, so the **growth factor** b is the ratio of the y-values: $b = \frac{66}{60} = \frac{11}{10} = 1.1$.
3. Use b and one data point to find the **initial value** a. Substitute 1.1 for b, 5 for x, and 66 for y:

$$66 = a(1.1)^5 \qquad a = \frac{66}{(1.1)^5} \approx 40.98$$

The function $y = 40.98(1.1)^x$ models the number of IM (in hundreds) Tia sends x years after 2018.

Common error. The growth factor $(1 + r)$ is different from the growth rate (r). Here the growth factor is 1.1 while the growth rate is 0.1, or 10%.

Check: $40.98(1.1)^4 \approx 60$. If the x-values are 2 apart, divide the y-values, then take the square root.

Say it: The x-values are consecutive, so the *growth factor* is $b =$ _____ $=$ _____ and the *growth rate* is _____%. To find the *initial value* a, I substitute _____ and solve, because ...

Try It 5: A model from two points

A Kansas realtor reported the value of an acre of farm land over a period of several years since 2015. The land was worth \$2010 in 2017 and \$2060 in 2018. Use the data to determine an exponential model that describes the value of the land per acre.

Hint: \$2010 is not the initial value. Use $x =$ years since 2015, find b, then use b and one point to solve for a.

Points: (____, 2010) and (____, 2060) $\quad b =$ _____ $\div$ _____ $\approx$ _____

_____ $= a\,($_____$)^{\square}$ $\quad a \approx$ _____ $\quad$ Model: $y \approx$ _____

Practice: A decrease per quarter

In 2022, a used car was worth \$18,500. Its value is decreasing at a rate of 12% per year.

Hint: a decrease of 12% per year keeps $1 - 0.12$ of the value each year, so the yearly factor is _____.

a. Complete the function that models the value y of the car, t years after 2022.

$y =$ _____ (_____)^t

b. To the nearest thousandth of a percent, the value is decreasing about _____% per quarter. Show the rewrite with exponent $4t$.

c. Why is the answer to part b not $12 \div 4 = 3\%$? Use the words *growth factor* and *Power of a Power rule*.

IXL: Practice for the quiz

Go to IXL, Algebra 2. Type the code in the search box. Do the starred skills first.

Code	Skill	Feeds
LWE	Evaluate exponential functions (warm-up)	Do Now: fractional powers on the calculator
E5S ★	Exponential growth and decay	Quiz 1 (Example 1): factor and rate
2YT ★	Exponential functions over unit intervals	Quiz 2 (Example 5): the ratio of y-values
YJW ★	Compound interest: word problems	Quiz 3 (bonus; Example 2, skipped here)

SmartScore 70 or above on each starred (★) skill replaces your quiz score.

Exit question: Use Structure (DOK 3)

The points $(2,\ 54.61)$ and $(4,\ 403.48)$ are points on the graph of an exponential model in the form $y = a \cdot e^x$.

Hint: $e \approx 2.718$ is the natural base; use the e^x key. Here the base is already known, so one point gives a and the other point checks it.

a. Find the exponential model.

b. Use the exponential model to find the value of y when $x = 8$.

Before the quiz, finish these sentences.

The *growth factor* is 1 plus the _____; for a decrease it is 1 minus the rate.

To change a yearly rate into a rate per quarter, I use the _____ to make the exponent _____, because ...

When two points have consecutive x-values, b is the _____ of their y-values; then I substitute one point to find the _____ a.